Triplet Sum Close to Target (medium)

Problem Statement

Given an array of unsorted numbers and a target number, find a **triplet in the array whose sum is as close to the target number as possible**, return the sum of the triplet. If there are more than one such triplet, return the sum of the triplet with the smallest sum.

Example 1:

```
Input: [-1, 0, 2, 3], target=3
Output: 2
Explanation: The triplet [-1, 0, 3] has the sum '2' which is closest to the target.

There are two triplets with distance '1' from the target: [-1, 0, 3] & [-1, 2, 3]. Between these two triplets, the correct answer will be [-1, 0, 3] as it has a sum '2' which is less than the sum of the other triplet which is '4'. This is because of the following requirement: 'If there are more than one such triplet, return the sum of the triplet with the smallest sum.'
```

Example 2:

```
Input: [-3, -1, 1, 2], target=1
Output: 0
Explanation: The triplet [-3, 1, 2] has the closest sum to the target.
```

Example 3:

```
Input: [1, 0, 1, 1], target=100
Output: 3
Explanation: The triplet [1, 1, 1] has the closest sum to the target.
```

Example 4:

```
Input: [0, 0, 1, 1, 2, 6], target=5
Output: 4
Explanation: There are two triplets with distance '1' from target: [1, 1, 2] & [0, 0, 6]. Between these two triplets, the correct answer will be [1, 1, 2] as it has a sum '4' which is less than the sum of the other triplet which is '6'. This is because of the following requirement: 'If there are more than one such triplet, return the sum of the triplet with the smallest sum.'
```

Try it yourself

Try solving this question here:

Java

```
import java.util.*;

class Solution {

  public int searchTriplet(int[] arr, int targetSum) {
    // TODO: Write your code here
    return 0;
  }
}
```

Test Case | Run Code Result | Stdout

```
[-1, 0, 2, 3]
3
```

Use Example Testcases | Run | Submit

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Solution: Triplet Sum To Zero

Next →
Solution: Triplet Sum Close To Target

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